



GQ RadonScan - Experimental Protocol Reference

Implementation profile for version 3.0.0

Document status

Experimental interoperable description of read mechanisms confirmed in real Re2.02 captures. Not an official GQ RFC.

Evidence levels: CONFIRMED / STRONGLY SUPPORTED / PROVISIONAL / UNKNOWN

Serial transport

- 115200 baud, 8N1, no handshake, DTR and RTS disabled.
- Treat replies as byte streams; SPIR replies have no separate header.
- Read each 4096-byte block exactly and with a timeout.

Confirmed commands

Command		Request bytes	Observed reply
	GETVER	3C 47 45 54 56 45 52 3E 3E	ASCII firmware, for example RadonScanRe2.02
	GETSERIAL	3C 47 45 54 53 45 52 49 41 4C 3E 3E	Observed 7-byte binary serial-number payload
	GETCFG	3C 47 45 54 43 46 47 3E 3E	Observed 512-byte configuration block
	GETCPM	3C 47 45 54 43 50 4D 3E 3E	Observed ASCII 0\r\n; not used as a radon value
	SPIR	3C 53 50 49 52 + A24 + L16 + 3E 3E	Raw read returning exactly L bytes

SPiR read frame

```
Request = '>'
```

Example: read 4096 bytes at address 0x1FC000

```
3C 53 50 49 52 1F C0 00 10 00 3E 3E
```

The reply contains exactly 4096 raw bytes. No checksum, header or terminator has been observed. Implementations must therefore use exact-length reads and reject short replies.

History structure

Address	Length	Observed role
0x1FC000	4096	Little-endian UInt32 CPH history; observed end offset 0x0CBC
0x1FD000	4096	Administration/diagnostic block; field at 0x270 remains provisional
0x1FE000	4096	14-byte records: AA 55 + seconds BE32 + 8 payload bytes
0x1FF000	4096	Configuration mirror; not required for hourly measurements

Reference algorithm

- 1. GETVER identifies RadonScanRe2.02.
- 2. Read 0x1FC000, 0x1FD000 and 0x1FE000.
- 3. Parse consecutive 14-byte AA55 records beginning at offset 0.
- 4. The t=0 record is a marker; raw count = time-record count - 1.
- 5. Read that number of little-endian UInt32 values backwards from the observed end offset 0x0CBC.
- 6. Pair raw values with time records 1..N.
- 7. Calculate $bq_m3 = raw_cph \times factor$.
- 8. Persist only hour indexes greater than the last stored index.

Validated test vectors

Capture	Hour index	CPH	Bq/m ³	Device
2026-07-25 06:34	36	1	1.530	1.5 displayed
2026-07-25 06:39	37	3	4.590	4.6 displayed
Earlier observation	n/a	2	3.060	3.1 displayed

Implementation requirements

- 115200 baud, 8N1, no handshake, DTR and RTS disabled.
- Treat replies as byte streams; SPIR replies have no separate header.
- Read each 4096-byte block exactly and with a timeout.
- Persist new hour indexes only and discard duplicates.
- Start a new measurement campaign when the hour index decreases.
- Store the conversion factor with every measurement.

Error handling

- Reject blocks whose length is not exactly 4096 bytes.
- Reject time records that are not separated by exactly 3600 seconds.
- Treat implausibly large CPH values as possible alignment errors.
- Do not publish a new measurement when the hour index is unchanged.
- Retain the last valid Home Assistant value while the USB device is temporarily unavailable.

Open questions

- Official meaning of the 0x1FD000 administration block.
- Behaviour after the 14-byte history area is full.
- Device-specific calibration parameters and exact internal rounding.
- Undocumented live-value commands for the current partial hour.

Security: this profile is read-only. Conforming implementations MUST NOT send SET, WRITE, ERASE or RESET commands.